THE MAKAR-LIMANOV INVARIANT EQUALS ITS SLICE VERSION IN DIMENSION THREE

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Abstract. Over any field of characteristic zero, the Makar-Limanov invariant of a domain coincides with its slice version whenever the latter has transcendence degree at most one or one less than that of the domain. In dimension three this settles a question of Dasgupta and Gupta, which had been answered only over algebraically closed fields.

Keywords: locally nilpotent derivation; Makar-Limanov invariant; slice.

MSC 2020: 14R10 (primary); 13N15, 13A50, 14R20.

1. Introduction

Let k be a field of characteristic zero and B a commutative k-domain. A k-derivation D of B is *locally nilpotent* if every element of B is killed by some power of D; the set of these derivations is written $\mathrm{LND}(B)$. Makar-Limanov introduced the ring

$$\mathrm{ML}(B) = \bigcap_{D \in \mathrm{LND}(B)} \ker D$$

to prove that the Koras–Russell threefold $x + x^2y + z^2 + t^3 = 0$ is not isomorphic to $\mathbb{C}^3$ although it is contractible [12]. Since then ML has been one of the main tools of affine algebraic geometry. Gupta used a characteristic-free version of it, built from exponential maps, to show that the Zariski cancellation problem has a negative answer for the affine three-space in positive characteristic [10, 11]. The invariant has a known weakness: it can become trivial after adding a variable. Dubouloz proved that the cylinder over the Koras–Russell threefold has trivial Makar-Limanov invariant, the same as the affine four-space [7].

Freudenburg introduced a variant that only uses derivations with a slice [8]. An element $s \in B$ is a *slice* of $D \in \mathrm{LND}(B)$ if $D(s) = 1$. Let $\mathrm{LND}^*(B)$ be the set of locally nilpotent derivations of B that have a slice, and put

$$\mathrm{ML}^*(B) = \bigcap_{D \in \mathrm{LND}^*(B)} \ker D, \qquad \text{with } \mathrm{ML}^*(B) = B \text{ if } \mathrm{LND}^*(B) = \emptyset.$$

If $\mathrm{LND}^*(B) \neq \emptyset$, the intersection defining $\mathrm{ML}(B)$ runs over more derivations, so $\mathrm{ML}(B) \subseteq \mathrm{ML}^*(B)$; if $\mathrm{LND}^*(B) = \emptyset$ the inclusion holds trivially. Dasgupta and Gupta used both rings to characterise the affine plane and the affine three-space [6]. Their main theorem says that, for a three-dimensional affine UFD B over an algebraically closed field of characteristic zero, the following are equivalent: (I) $B = k^{[3]}$; (II) $\mathrm{ML}^*(B) = k$; (III) $\mathrm{ML}(B) = k$ and $\mathrm{ML}^*(B) \neq B$. They showed that the hypotheses "algebraically closed" and "UFD" cannot be dropped from (III)⇒(I), and they asked whether they can be dropped from (III)⇒(II).[1]

Question (Dasgupta–Gupta [6])**.** Does there exist a three-dimensional affine k-domain B over a field k of characteristic zero such that $\mathrm{ML}(B) = k$ but $\mathrm{tr.deg}_k \mathrm{ML}^*(B) = 1$?

[1]The question is numbered 4.8 in the preprint version of [6] and is cited as Question 5.9 in [9]. We quote it verbatim.

They proved that the answer is negative when k is algebraically closed and B is a UFD, and observed that a negative answer in general would give (III)⇒(II) for every three-dimensional affine domain over every field of characteristic zero. Gaifullin and Shafarevich then proved that $\mathrm{ML}(B) = \mathrm{ML}^*(B)$ whenever $\mathrm{LND}^*(B) \neq \emptyset$, for every affine domain over an *algebraically closed* field of characteristic zero [9, Theorem 1]. Their proof compares orbits of the subgroups of $\mathrm{Aut}(B)$ generated by the corresponding unipotent one-parameter groups, through the work of Arzhantsev, Flenner, Kaliman, Kutzschebauch and Zaidenberg on flexible varieties [1]. It needs points over k in general position and so does not apply to fields such as $\mathbb{R}$ or $\mathbb{Q}$. Neither invariant commutes with extension of the base field (Section 5), so the question over non-closed fields cannot be reduced to the closed case.

We answer the question over every field of characteristic zero, without any finiteness or factoriality assumption, by a short argument with no geometry in it.

Theorem A. *Let k be a field of characteristic zero and B a k-domain with $\mathrm{tr.deg}_k \mathrm{ML}^*(B) \leq 1$. Then $\mathrm{ML}(B) = \mathrm{ML}^*(B)$.*

Theorem B. *Let k be a field of characteristic zero and B a k-domain of finite transcendence degree $n \geq 1$ over k with $\mathrm{tr.deg}_k \mathrm{ML}^*(B) = n - 1$. Then $\mathrm{ML}(B) = \mathrm{ML}^*(B) = \ker E$ for every nonzero $E \in \mathrm{LND}(B)$, the ring $\mathrm{ML}^*(B)$ has no nonzero locally nilpotent derivation, and B is a polynomial ring in one variable over $\mathrm{ML}^*(B)$.*

For affine domains Theorem B is a lemma of Dasgupta and Gupta [6, Lemma 3.2], whose proof rests on a theorem of Crachiola and Makar-Limanov on rigid rings [5]. The proof given here does not use that theorem, and it works for every domain of finite transcendence degree. For $\mathrm{tr.deg}_k B \leq 2$ and $\mathrm{LND}^*(B) \neq \emptyset$ the equality $\mathrm{ML}(B) = \mathrm{ML}^*(B)$ was known over every field of characteristic zero [6, 13]; Theorem A is new over fields that are not algebraically closed as soon as $\mathrm{tr.deg}_k B \geq 3$. Together they cover every value of $\mathrm{tr.deg}_k \mathrm{ML}^*(B)$ when $\mathrm{tr.deg}_k B \leq 3$.

Theorem C. *Let k be a field of characteristic zero and B a k-domain with $\mathrm{tr.deg}_k B \leq 3$ and $\mathrm{LND}^*(B) \neq \emptyset$. Then $\mathrm{ML}(B) = \mathrm{ML}^*(B)$.*

Corollary D. *Over a field of characteristic zero there is no domain B of transcendence degree three, affine or not, with $\mathrm{ML}(B) = k$ and $\mathrm{tr.deg}_k \mathrm{ML}^*(B) = 1$. More generally, if $\mathrm{tr.deg}_k B = n$, $\mathrm{ML}(B) = k$ and $k \neq \mathrm{ML}^*(B) \neq B$, then $2 \leq \mathrm{tr.deg}_k \mathrm{ML}^*(B) \leq n - 2$. In particular the conditions (II) and (III) above are equivalent for every three-dimensional domain over every field of characteristic zero.*

The proof rests on one observation, which has nothing to do with the dimension: $\mathrm{ML}^*(B)$ is mapped into itself by *every* locally nilpotent derivation of B, not only by those with a slice (Proposition 3.2). The reason is that the exponential of such a derivation is an automorphism of B, and conjugating a derivation with a slice by an automorphism gives another derivation with a slice. Once this is known, any locally nilpotent derivation E restricts to $\mathrm{ML}^*(B)$, and if the restriction is not zero, a local slice of the restriction can be divided by its image when the kernel of the restriction is small. The quotient is a slice of E itself, so E kills $\mathrm{ML}^*(B)$ after all. This works when the kernel of the restriction is algebraic over k, which is guaranteed by $\mathrm{tr.deg}_k \mathrm{ML}^*(B) \leq 1$. In the case $\mathrm{tr.deg}_k \mathrm{ML}^*(B) = n - 1$ the ring $\mathrm{ML}^*(B)$ is the kernel of a slice derivation, and it has no nonzero locally nilpotent derivations at all.

The remaining range $2 \leq \mathrm{tr.deg}_k \mathrm{ML}^*(B) \leq n - 2$ for $n \geq 4$ is open over non-closed fields; Figure 1 shows what is settled.

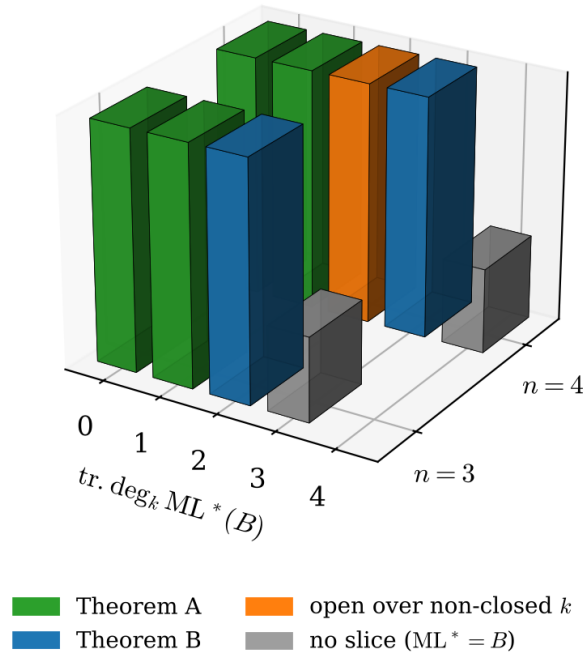


FIGURE 1. Which values of $\operatorname{tr.deg}_k \mathrm{ML}^*(B)$ are settled for $\operatorname{tr.deg}_k B = 3$ and $\operatorname{tr.deg}_k B = 4$ over an arbitrary field of characteristic zero. The value $\operatorname{tr.deg}_k B$ itself occurs exactly when $\mathrm{LND}^*(B) = \emptyset$, where the two rings may differ [6]. Over algebraically closed fields every value is settled by [9].

The paper is self-contained apart from elementary facts on transcendence degree. Section 2 proves the facts on locally nilpotent derivations that we use, including the slice theorem. Section 3 contains the stability of ML^*. Section 4 proves the theorems, and Section 5 discusses examples over the real numbers which show that the two invariants change under extension of scalars.

Conventions. Throughout, k is a field of characteristic zero, rings are commutative with identity, and B is a k-domain, that is, a k-algebra which is an integral domain. We write $\operatorname{tr.deg}_k C$ for the transcendence degree over k of the fraction field of a k-domain C. An element b of B is *algebraic over* a subring C if $P(b) = 0$ for a nonzero polynomial $P \in C[T]$; since denominators can be cleared, this is the same as being algebraic over $\mathrm{Frac}(C)$. A subring $C \subseteq B$ is *algebraically closed in* B if every element of B algebraic over C lies in C. For a ring C, $C^{[1]}$ denotes a polynomial ring in one variable over C. A domain is *rigid* if it has no nonzero locally nilpotent derivation.

2. Locally nilpotent derivations

A k-derivation D of B is k-linear and satisfies $D(ab) = aD(b) + bD(a)$. From $D(1) = D(1 \cdot 1) = 2D(1)$ we get $D(1) = 0$, hence $D(\lambda) = 0$ for $\lambda \in k$, and $\ker D$ is a k-subalgebra of

B. By induction on m one has the Leibniz formula

$$D^m(ab) = \sum_{i=0}^{m} \binom{m}{i} D^i(a)\, D^{m-i}(b). \tag{1}$$

Lemma 2.1. *Let D be a k-derivation of B. The elements of B killed by some power of D form a k-subalgebra. In particular, if B is generated as a k-algebra by elements each killed by a power of D, then $D \in \mathrm{LND}(B)$.*

Proof. The set is closed under sums and multiplication by elements of k, since D^m is k-linear and $D^m(b) = 0$ implies $D^{m'}(b) = 0$ for $m' \geq m$. If $D^m(a) = 0$ and $D^n(b) = 0$, every term of (1) for $D^{m+n}(ab)$ vanishes: for $i \geq m$ the factor $D^i(a)$ is zero, and for $i < m$ the factor $D^{m+n-i}(b)$ is zero because $m + n - i > n$. □

For $D \in \mathrm{LND}(B)$ and $t \in k$ put

$$\exp(tD)(b) = \sum_{i \geq 0} \frac{t^i}{i!} D^i(b),$$

a finite sum for each b. Division by $i!$ is allowed because k has characteristic zero.

Lemma 2.2. *For $D \in \mathrm{LND}(B)$ and $t, u \in k$, the map $\exp(tD)$ is a k-algebra endomorphism of B, and $\exp(tD) \circ \exp(uD) = \exp((t+u)D)$. In particular $\exp(tD)$ is an automorphism with inverse $\exp(-tD)$.*

Proof. Linearity over k and $\exp(tD)(1) = 1$ are clear. By (1),

$$\exp(tD)(ab) = \sum_{m \geq 0} \frac{t^m}{m!} \sum_{i+j=m} \frac{m!}{i!\, j!} D^i(a) D^j(b) = \Big(\sum_i \frac{t^i}{i!} D^i(a)\Big)\Big(\sum_j \frac{t^j}{j!} D^j(b)\Big).$$

Since $\exp(uD)(b)$ is a finite k-linear combination of the $D^j(b)$, all sums below are finite, and

$$\exp(tD)\exp(uD)(b) = \sum_{i,j} \frac{t^i u^j}{i!\, j!} D^{i+j}(b) = \sum_m \frac{1}{m!}\Big(\sum_{i+j=m} \frac{m!}{i!\, j!} t^i u^j\Big) D^m(b) = \exp((t+u)D)(b).$$

Taking $u = -t$ and using $\exp(0 \cdot D) = \mathrm{id}$ gives the inverse. □

The next lemma is the place where characteristic zero enters most directly.

Lemma 2.3. *Let δ be a k-derivation of a k-domain C and $A = \ker \delta$. Every element of C algebraic over A lies in A. In particular $\ker \delta$ is algebraically closed in C, and every element of C algebraic over k lies in $\ker \delta$.*

Proof. Let $b \in C$ be algebraic over A, and choose a nonzero $P = \sum_{i=0}^{m} a_i T^i \in A[T]$ with $P(b) = 0$ and m as small as possible; then $m \geq 1$ and $a_m \neq 0$, since a nonzero constant does not vanish at b. Applying δ to $P(b) = 0$ and using $\delta(a_i) = 0$ gives

$$0 = \sum_{i=1}^{m} i a_i b^{i-1} \delta(b) = P'(b)\, \delta(b), \qquad P' = \sum_{i=1}^{m} i a_i T^{i-1} \in A[T].$$

The leading coefficient $m a_m$ of P' is nonzero because k has characteristic zero and C is a domain, so P' is a nonzero polynomial of degree $m - 1 < m$. By the minimality of m, $P'(b) \neq 0$. As C is a domain, $\delta(b) = 0$. The last assertion follows because $k \subseteq A$. □

Lemma 2.4. *Let δ be a nonzero locally nilpotent derivation of a k-domain C and $A = \ker \delta$. There is $g \in C$ with $c = \delta(g) \neq 0$ and $\delta(c) = 0$. Such an element g is not algebraic over A. Consequently, if $\mathrm{tr.deg}_k C$ is finite, then $\mathrm{tr.deg}_k A \leq \mathrm{tr.deg}_k C - 1$.*

Proof. Pick $f \in C$ with $\delta(f) \neq 0$ and let $N \geq 1$ be the largest integer with $\delta^N(f) \neq 0$; it exists because δ is locally nilpotent. Then $g = \delta^{N-1}(f)$ has $\delta(g) = \delta^N(f) \neq 0$ and $\delta^2(g) = 0$. If g were algebraic over A, Lemma 2.3 would give $\delta(g) = 0$. For the last claim, let $x_1, \ldots, x_r \in A$ be algebraically independent over k. As g is not algebraic over $k[x_1, \ldots, x_r] \subseteq A$, the elements $x_1, \ldots, x_r, g$ of C are algebraically independent, so $r + 1 \leq \operatorname{tr.deg}_k C$. □

An element g as in Lemma 2.4 is called a *local slice* of δ. The next result is the slice theorem; we include the short proof.[2]

Lemma 2.5 (Slice theorem)**.** *Let $D \in \mathrm{LND}(B)$ have a slice s, and let $A = \ker D$. Then s is not algebraic over A, and $B = A[s]$. Thus $B = A^{[1]}$, the derivation D is the derivative d/ds with respect to s, and if $\operatorname{tr.deg}_k B = n$ is finite then $\operatorname{tr.deg}_k A = n - 1$.*

Proof. Since $D(s) = 1 \neq 0$, Lemma 2.3 shows that s is not algebraic over A, so the A-algebra map $A[T] \to B$, $T \mapsto s$, is injective. For $b \in B$ put

$$\pi(b) = \sum_{i \geq 0} \frac{(-1)^i}{i!} D^i(b)\, s^i,$$

a finite sum. Since $D(s) = 1$,

$$D(\pi(b)) = \sum_{i \geq 0} \frac{(-1)^i}{i!} D^{i+1}(b) s^i + \sum_{i \geq 1} \frac{(-1)^i}{(i-1)!} D^i(b) s^{i-1}.$$

After the substitution $i \mapsto i + 1$ the second sum is $\sum_{i \geq 0} \frac{(-1)^{i+1}}{i!} D^{i+1}(b) s^i$, which cancels the first. Hence $\pi(b) \in A$ for every b. Now

$$\sum_{j \geq 0} \frac{1}{j!} \pi\big(D^j(b)\big) s^j = \sum_{i,j \geq 0} \frac{(-1)^i}{i!\, j!} D^{i+j}(b) s^{i+j} = \sum_{m \geq 0} \frac{D^m(b) s^m}{m!} \sum_{i=0}^{m} \binom{m}{i} (-1)^i = b,$$

since the inner sum is $(1-1)^m$, which is 0 for $m \geq 1$ and 1 for $m = 0$. Each $\pi(D^j(b))$ lies in A, so $b \in A[s]$. On $A[s]$ the derivation D kills A and sends s to 1, so it agrees with d/ds. Finally $\mathrm{Frac}(B) = \mathrm{Frac}(A)(s)$ with s transcendental over $\mathrm{Frac}(A)$, so $\operatorname{tr.deg}_k B = \operatorname{tr.deg}_k A + 1$. □

Lemma 2.6. *If $D_1, D_2 \in \mathrm{LND}(B)$ commute, then $D_1 + D_2 \in \mathrm{LND}(B)$.*

Proof. Since $D_1 D_2 = D_2 D_1$, the binomial formula gives $(D_1 + D_2)^N = \sum_{j=0}^{N} \binom{N}{j} D_1^{N-j} D_2^j$. Fix $b \in B$. Choose p with $D_2^p(b) = 0$, and then q with $D_1^q(D_2^j(b)) = 0$ for the finitely many $j < p$. For $N \geq p + q$ every term vanishes: if $j \geq p$ then $D_2^j(b) = 0$, and if $j < p$ then $N - j > q$ and $D_1^{N-j} D_2^j(b) = 0$. □

3. Stability of the slice invariant

From now on write $R = \mathrm{ML}^*(B)$. Each kernel $\ker D$ is a k-subalgebra of B, so R is a k-subalgebra. If $\mathrm{LND}^*(B) \neq \emptyset$, R is algebraically closed in B: an element algebraic over R is algebraic over each $\ker D$, $D \in \mathrm{LND}^*(B)$, so it lies in each of them by Lemma 2.3. The same holds trivially when $R = B$.

Lemma 3.1. *For every k-algebra automorphism σ of B one has $\sigma(R) = R$.*

[2]The map π below is the Dixmier map of s; formally it is $\exp(-sD)$. For the slice theorem in the literature see [8, Chapter 1].

Proof. If $\mathrm{LND}^*(B) = \emptyset$, then $R = B$ and there is nothing to prove. Let $D \in \mathrm{LND}^*(B)$ with slice s, and put $D^\sigma = \sigma D \sigma^{-1}$. It is a k-derivation:

$$D^\sigma(ab) = \sigma\big(\sigma^{-1}(a)D(\sigma^{-1}b) + \sigma^{-1}(b)D(\sigma^{-1}a)\big) = aD^\sigma(b) + bD^\sigma(a).$$

It is locally nilpotent since $(D^\sigma)^m = \sigma D^m \sigma^{-1}$, it has the slice $\sigma(s)$ because $D^\sigma(\sigma(s)) = \sigma(D(s)) = \sigma(1) = 1$, and $\ker D^\sigma = \sigma(\ker D)$. The map $D \mapsto D^\sigma$ is a bijection of $\mathrm{LND}^*(B)$ onto itself, with inverse $D \mapsto D^{\sigma^{-1}}$. Hence

$$\sigma(R) = \bigcap_{D \in \mathrm{LND}^*(B)} \sigma(\ker D) = \bigcap_{D \in \mathrm{LND}^*(B)} \ker D^\sigma = R. \qquad \square$$

Proposition 3.2. *Every $E \in \mathrm{LND}(B)$ satisfies $E(R) \subseteq R$. Hence E restricts to a locally nilpotent derivation $E|_R$ of R.*

Proof. Let $a \in R$ and choose N with $E^{N+1}(a) = 0$. For $t \in k$, Lemmas 2.2 and 3.1 give

$$\exp(tE)(a) = \sum_{i=0}^{N} t^i \, \frac{E^i(a)}{i!} \in R. \tag{2}$$

The field k contains $\mathbb{Q}$ and is infinite, so we may choose distinct $t_0, \dots, t_N \in k$. The Vandermonde matrix $V = (t_j^i)_{0 \le j,i \le N}$ has determinant $\prod_{j<j'}(t_{j'} - t_j) \neq 0$, so it is invertible over k. By (2) the column vector with entries $E^i(a)/i!$ equals V^{-1} applied to the column with entries $\exp(t_j E)(a) \in R$. Its entries are therefore k-linear combinations of elements of R, and lie in R because R is a k-subspace. In particular $E(a) \in R$. The restriction is locally nilpotent because E is. $\square$

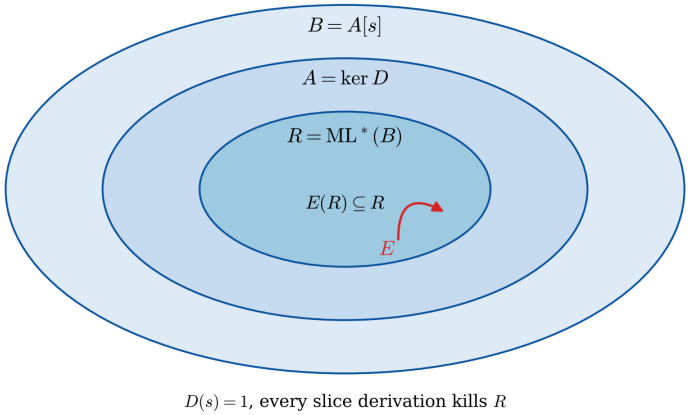


FIGURE 2. The rings in the proofs. A slice derivation D writes B as $A[s]$ with $A = \ker D$, and $R = \mathrm{ML}^*(B)$ lies in every such A. By Proposition 3.2 an arbitrary locally nilpotent derivation E maps R into itself, even when E moves A.

The following extension lemma is [6, Lemma 3.1(i)]; we recall the proof, which we also need for non-affine rings.

Lemma 3.3. *Let $D \in \mathrm{LND}^*(B)$ and $A = \ker D$. Then $R \subseteq \mathrm{ML}(A)$.*

Proof. By Lemma 2.5, $B = A[s]$ is a polynomial ring and $D = d/ds$. Let $\delta \in \mathrm{LND}(A)$ and define $\tilde{\delta}$ on $A[s]$ coefficientwise, $\tilde{\delta}(\sum_i a_i s^i) = \sum_i \delta(a_i)s^i$. It is a k-derivation, since δ is one on A and $\tilde{\delta}(s) = 0$. It is locally nilpotent, because $\tilde{\delta}^m(\sum_i a_i s^i) = \sum_i \delta^m(a_i)s^i$ and only finitely many a_i occur. It commutes with D, since for $a \in A$

$$\tilde{\delta}D(as^i) = \tilde{\delta}(ia\, s^{i-1}) = i\,\delta(a)s^{i-1} = D\big(\delta(a)s^i\big) = D\tilde{\delta}(as^i).$$

By Lemma 2.6, $E = \tilde{\delta} + D$ is locally nilpotent, and $E(s) = 0 + 1 = 1$, so $E \in \mathrm{LND}^*(B)$. For $a \in A$ we have $E(a) = \delta(a)$, so $\ker E \cap A = \ker \delta$. Since $R \subseteq \ker D = A$ and $R \subseteq \ker E$, we get $R \subseteq \ker \delta$. This holds for every $\delta \in \mathrm{LND}(A)$, so $R \subseteq \mathrm{ML}(A)$. □

Remark 3.4. Lemma 3.1 and Proposition 3.2 say that R is stable under the whole automorphism group of B and under every locally nilpotent derivation, although it is defined by a smaller family. The ring $\mathrm{ML}(B)$ is the ring of invariants of the subgroup of $\mathrm{Aut}(B)$ generated by all the groups $\{\exp(tD) : t \in k\}$, and R is the ring of invariants of the subgroup generated by those with $D \in \mathrm{LND}^*(B)$; the second subgroup is normal in the first. Over an algebraically closed field Gaifullin and Shafarevich show that the two subgroups have the same general orbits [9]. Group actions and the structures they preserve in other geometric settings are studied in [2, 3, 4].

4. Proofs of the theorems

Proof of Theorem A. We have $\mathrm{ML}(B) \subseteq R$ (Section 1), so it suffices to show that $R \subseteq \ker E$ for every $E \in \mathrm{LND}(B)$. Fix E and let $\varepsilon = E|_R$, which is a locally nilpotent derivation of the domain R by Proposition 3.2. Suppose, to reach a contradiction, that $\varepsilon \neq 0$, and let $K = \ker \varepsilon \subseteq R$.

By Lemma 2.4 there is $g \in R$ with $c = \varepsilon(g) \neq 0$ and $\varepsilon(c) = 0$, and g is not algebraic over K. We claim that every element of K is algebraic over k. Otherwise some $x \in K$ is transcendental over k; as g is not algebraic over $k[x] \subseteq K$, the elements x and g of R are algebraically independent over k, which contradicts $\mathrm{tr.deg}_k R \leq 1$.

So $c \in K$ is a nonzero element algebraic over k. Let $\mu(T) = T^m + \lambda_{m-1}T^{m-1} + \cdots + \lambda_0 \in k[T]$ be a monic polynomial of least degree with $\mu(c) = 0$. If $\lambda_0 = 0$, then $c \cdot (c^{m-1} + \lambda_{m-1}c^{m-2} + \cdots + \lambda_1) = 0$ with $c \neq 0$ in a domain, which contradicts the minimality of m. Hence $\lambda_0 \neq 0$ and

$$c^{-1} = -\lambda_0^{-1}\big(c^{m-1} + \lambda_{m-1}c^{m-2} + \cdots + \lambda_1\big) \in k[c] \subseteq K.$$

Put $s = c^{-1}g \in R \subseteq B$. Since $c^{-1} \in K$, we have $E(c^{-1}) = \varepsilon(c^{-1}) = 0$, and therefore

$$E(s) = E(c^{-1})\, g + c^{-1}E(g) = c^{-1}c = 1.$$

Thus s is a slice of E, so $E \in \mathrm{LND}^*(B)$, and by the definition of R we get $R \subseteq \ker E$, that is, $\varepsilon = 0$. This contradiction shows $\varepsilon = 0$, so $R \subseteq \ker E$. □

The argument shows a little more: if $E \in \mathrm{LND}(B)$ does not kill $\mathrm{ML}^*(B)$, then $\ker(E|_{\mathrm{ML}^*(B)})$ contains an element transcendental over k.[3] The step that fails in general is the inversion of c: for $\mathrm{tr.deg}_k \mathrm{ML}^*(B) \geq 2$ the element $c = \varepsilon(g)$ may be a non-unit of K, and $c^{-1}g$ is then not in B.

Proof of Theorem B. Since $\mathrm{tr.deg}_k R = n-1 < n = \mathrm{tr.deg}_k B$, we have $R \neq B$, so $\mathrm{LND}^*(B) \neq \emptyset$. Fix $D \in \mathrm{LND}^*(B)$ with slice s and put $A = \ker D$. By Lemma 2.5, $B = A[s] = A^{[1]}$ and $\mathrm{tr.deg}_k A = n - 1$.

[3] When $\mathrm{tr.deg}_k \mathrm{ML}^*(B) = 0$ the theorem is also immediate from Lemma 2.3: every element of $\mathrm{ML}^*(B)$ is then algebraic over k, hence killed by every derivation of B.

Step 1: $R = A$. We have $R \subseteq A$. Choose $x_1, \dots, x_{n-1} \in R$ algebraically independent over k; this is possible because $\mathrm{Frac}(R)$ is generated by R. Let $a \in A$. If a were not algebraic over $k[x_1, \dots, x_{n-1}]$, the n elements $x_1, \dots, x_{n-1}, a$ of A would be algebraically independent, against $\mathrm{tr.deg}_k A = n - 1$. So a is algebraic over R, and $a \in R$ because R is algebraically closed in B (Section 3). Hence $A = R$, and $B = R^{[1]}$.

Step 2: R is rigid. By Lemma 3.3 and Step 1, $R \subseteq \mathrm{ML}(A) = \mathrm{ML}(R)$. So every $\delta \in \mathrm{LND}(R)$ vanishes on R, that is, $\mathrm{LND}(R) = \{0\}$.

Step 3: $\mathrm{ML}(B) = R$. Let $E \in \mathrm{LND}(B)$. By Proposition 3.2, $E|_R \in \mathrm{LND}(R) = \{0\}$, so $R \subseteq \ker E$. Hence $R \subseteq \mathrm{ML}(B) \subseteq R$.

Step 4: $\ker E = R$ *for every nonzero* $E \in \mathrm{LND}(B)$. By Step 3, $R \subseteq \ker E$, and by Lemma 2.4, $\mathrm{tr.deg}_k \ker E \leq n - 1 = \mathrm{tr.deg}_k R$. As in Step 1, every element of $\ker E$ is algebraic over R and hence lies in R. □

Step 4 says that B is semi-rigid: all nonzero locally nilpotent derivations have the same kernel.

Proof of Theorem C. Let $n = \mathrm{tr.deg}_k B$. Fix $D \in \mathrm{LND}^*(B)$ with slice s and $A = \ker D$. Since s is transcendental over k by Lemma 2.5, $n \geq 1$, and $\mathrm{tr.deg}_k A = n - 1 \leq 2$. As $R \subseteq A$, $\mathrm{tr.deg}_k R \leq n - 1$. If $\mathrm{tr.deg}_k R \leq 1$, Theorem A applies. The only remaining case is $n = 3$ and $\mathrm{tr.deg}_k R = 2 = n - 1$, where Theorem B applies. □

Proof of Corollary D. Let $\mathrm{tr.deg}_k B = n$, $\mathrm{ML}(B) = k$ and $k \neq R \neq B$. Since $R \neq B$, $\mathrm{LND}^*(B) \neq \emptyset$ and $\mathrm{tr.deg}_k R \leq n - 1$ by Lemma 2.5. If $\mathrm{tr.deg}_k R \leq 1$ or $\mathrm{tr.deg}_k R = n - 1$, Theorems A and B give $R = \mathrm{ML}(B) = k$, which is excluded. Hence $2 \leq \mathrm{tr.deg}_k R \leq n - 2$. For $n = 3$ this range is empty, and since $\mathrm{tr.deg}_k R = 1$ forces $R \neq k$ and $R \neq B$, no three-dimensional B has $\mathrm{ML}(B) = k$ and $\mathrm{tr.deg}_k R = 1$.

For the equivalence, let $\mathrm{tr.deg}_k B = 3$. If (III) holds, that is $\mathrm{ML}(B) = k$ and $R \neq B$, the case $R \neq k$ has just been excluded, so $R = k$, which is (II). If (II) holds, then $R = k \neq B$ because $\mathrm{tr.deg}_k B = 3$, and $k \subseteq \mathrm{ML}(B) \subseteq R = k$, which is (III). □

Remark 4.1. The implication (III)⇒(I) of [6] is false without its hypotheses: for $k = \mathbb{R}$ the ring $B = \mathbb{R}[X, Y, Z]/(XY - Z^2 - 1)\,[T]$ satisfies $\mathrm{ML}^*(B) = \mathbb{R}$ but is not $\mathbb{R}^{[3]}$ [6]. Corollary D shows that the failure happens only between (I) and the other two conditions.

5. Examples over the real numbers

The examples in this section show that ML and ML* do not commute with extension of the base field, so that results over $\mathbb{C}$ cannot simply be descended to $\mathbb{R}$. They also show that the hypotheses of Theorems A and B occur for rings that are not polynomial rings over k.

Let $S = \mathbb{R}[x, y, z] = \mathbb{R}[X, Y, Z]/(X^2 + Y^2 + Z^2 - 1)$ be the coordinate ring of the real two-sphere, and $S_{\mathbb{C}} = S \otimes_{\mathbb{R}} \mathbb{C}$. The polynomial $X^2 + Y^2 + Z^2 - 1$ is irreducible over $\mathbb{C}$, so S and $S_{\mathbb{C}}$ are domains of transcendence degree two.[4]

Proposition 5.1. (a) *S is rigid over $\mathbb{R}$, so* $\mathrm{ML}(S) = \mathrm{ML}^*(S) = S$.
(b) $\mathrm{ML}(S[T]) = \mathrm{ML}^*(S[T]) = S$, *which has transcendence degree two over* $\mathbb{R}$.
(c) $\mathrm{ML}(S_{\mathbb{C}}) = \mathbb{C}$, *and* $\mathrm{ML}(S_{\mathbb{C}}[T]) = \mathrm{ML}^*(S_{\mathbb{C}}[T]) = \mathbb{C}$.

[4]If it factored into two linear polynomials, its homogenisation $X^2 + Y^2 + Z^2 - W^2$ would be a product of two linear forms; but a product of two linear forms is a quadratic form of rank at most two, and this one has rank four. The transcendence degree is two because x, y are algebraically independent (no nonzero polynomial in X, Y alone lies in the ideal) and z is algebraic over them.

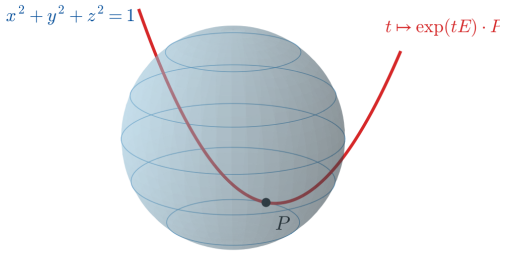


FIGURE 3. The proof of Proposition 5.1(a). For a real point P the orbit $t \mapsto \exp(tE)\cdot P$ is a polynomial curve which must stay on the sphere; a nonconstant real polynomial curve is unbounded, as drawn, so the orbit is a single point.

Proof. (a) A point $P = (p_1, p_2, p_3) \in \mathbb{R}^3$ with $p_1^2 + p_2^2 + p_3^2 = 1$ defines an $\mathbb{R}$-algebra homomorphism $\mathrm{ev}_P \colon S \to \mathbb{R}$. Let $E \in \mathrm{LND}(S)$. For $f \in S$ the function $t \mapsto \mathrm{ev}_P(\exp(tE)f) = \sum_i t^i \,\mathrm{ev}_P(E^i f)/i!$ is a real polynomial in t. Let $X(t), Y(t), Z(t)$ be these polynomials for $f = x, y, z$. Since $\exp(tE)$ is an algebra homomorphism (Lemma 2.2),

$$X(t)^2 + Y(t)^2 + Z(t)^2 = \mathrm{ev}_P\big(\exp(tE)(x^2+y^2+z^2)\big) = \mathrm{ev}_P(1) = 1$$

for all $t \in \mathbb{R}$. If the largest of the degrees of X, Y, Z were $d \geq 1$, the coefficient of t^{2d} on the left would be the sum of the squares of the leading coefficients of those polynomials of degree d, which is positive. So X, Y, Z are constant, and comparing the coefficients of t gives $\mathrm{ev}_P(Ex) = \mathrm{ev}_P(Ey) = \mathrm{ev}_P(Ez) = 0$.

Thus Ex, Ey, Ez vanish at every real point of the sphere. We show that such an element $f \in S$ is zero. Dividing by the monic polynomial $Z^2 - (1 - X^2 - Y^2)$ in Z, write $f = a(x,y) + z\,b(x,y)$ with $a, b \in \mathbb{R}[X, Y]$. For (p_1, p_2) in the open unit disc, both $(p_1, p_2, \pm\sqrt{1 - p_1^2 - p_2^2})$ are real points, so $a(p_1, p_2) \pm r\, b(p_1, p_2) = 0$ with $r = \sqrt{1 - p_1^2 - p_2^2} > 0$. Hence a and b vanish on the open disc, so $a = b = 0$ as polynomials, and $f = 0$. Therefore E kills the generators x, y, z, so $E = 0$. Since $\mathrm{LND}(S) = \{0\}$, also $\mathrm{LND}^*(S) = \emptyset$ and $\mathrm{ML}^*(S) = S$.

(b) The derivative ∂_T has the slice T and kernel S, so $\mathrm{ML}(S[T]) \subseteq \mathrm{ML}^*(S[T]) \subseteq S$. Conversely, let $E \in \mathrm{LND}(S[T])$. The argument of (a), applied at the real points (P, τ) with P on the sphere and $\tau \in \mathbb{R}$, shows that Ex, Ey, Ez vanish at all of them. An element $\sum_j f_j T^j$ of $S[T]$ ($f_j \in S$) vanishing at all these points has $f_j(P) = 0$ for all j and all P, since a real polynomial in τ with infinitely many roots is zero; so each $f_j = 0$ by (a). Hence E kills S, and $S \subseteq \mathrm{ML}(S[T])$. The same conclusion follows from Theorem B, once one knows that $\mathrm{tr.deg}_k \mathrm{ML}^*(S[T]) = 2$.

(c) Put $u = x + iy$ and $v = x - iy$ in $S_{\mathbb{C}}$. Then $uv = x^2 + y^2 = 1 - z^2$, and $S_{\mathbb{C}} = \mathbb{C}[u, v, z]$ with the single relation $uv + z^2 - 1 = 0$. Define a $\mathbb{C}$-derivation D_1 by $D_1(u) = 0$, $D_1(z) = u$, $D_1(v) = -2z$. It is well defined, since $D_1(uv + z^2 - 1) = u(-2z) + 2zu = 0$. It is locally nilpotent

by Lemma 2.1, since $D_1^2(z) = 0$ and $D_1^3(v) = -2D_1^2(z) = 0$. Symmetrically, $D_2(v) = 0$, $D_2(z) = v$, $D_2(u) = -2z$ defines $D_2 \in \mathrm{LND}(S_{\mathbb{C}})$.

The localisation $S_{\mathbb{C}}[u^{-1}]$ equals $\mathbb{C}[u^{\pm 1}, z]$, because $v = (1 - z^2)/u$. Here u and z are algebraically independent, since they generate a field of transcendence degree two. Extend D_1 to $S_{\mathbb{C}}[u^{-1}]$ by $D_1(b/u^m) = D_1(b)/u^m$, which is possible as $D_1(u) = 0$. On $\mathbb{C}[u^{\pm 1}][z]$ it acts as $u\,\partial/\partial z$, whose kernel is $\mathbb{C}[u^{\pm 1}]$. Hence $\ker D_1 \subseteq \mathbb{C}(u)$, and in the same way $\ker D_2 \subseteq \mathbb{C}(v)$. The elements u, v are algebraically independent, because z is algebraic over $\mathbb{C}(u, v)$ through $z^2 = 1 - uv$. In the field $\mathbb{C}(u, v)$ one has $\mathbb{C}(u) \cap \mathbb{C}(v) = \mathbb{C}$.[5] So $\mathrm{ML}(S_{\mathbb{C}}) \subseteq \ker D_1 \cap \ker D_2 = \mathbb{C}$.

Extend D_1 and D_2 to $S_{\mathbb{C}}[T]$ by $D_i(T) = 0$. Each commutes with ∂_T, so $D_i + \partial_T$ is locally nilpotent by Lemma 2.6, and it has the slice T. On $S_{\mathbb{C}}$ it agrees with D_i. Using also ∂_T, whose kernel is $S_{\mathbb{C}}$,

$$\mathrm{ML}^*(S_{\mathbb{C}}[T]) \subseteq S_{\mathbb{C}} \cap \ker(D_1 + \partial_T) \cap \ker(D_2 + \partial_T) = \ker D_1 \cap \ker D_2 = \mathbb{C}.$$

Since $\mathrm{ML} \subseteq \mathrm{ML}^*$ and both contain $\mathbb{C}$, both equal $\mathbb{C}$. □

So the real ring $S[T]$ has $\mathrm{ML} = \mathrm{ML}^* = S$, of transcendence degree two, while after extension to $\mathbb{C}$ both invariants become $\mathbb{C}$. The equality $\mathrm{ML} = \mathrm{ML}^*$ over $\mathbb{C}$, which is part of [9, Theorem 1], therefore carries no information about the two real invariants: both change. The hyperboloid used by Dasgupta and Gupta behaves in the opposite way, with ML^* already trivial over $\mathbb{R}$ [6].

Example 5.2. Let $C = \mathbb{R}[x, y]/(x^2 + y^2 - 1)$ be the coordinate ring of the real circle and $B = C[T_1, T_2]$. The argument of Proposition 5.1(a), with $X(t)^2 + Y(t)^2 = 1$, shows that C is rigid and that every locally nilpotent derivation of B kills C. The derivations ∂_{T_1} and ∂_{T_2} have slices and $\ker \partial_{T_1} \cap \ker \partial_{T_2} = C$. Hence $\mathrm{ML}(B) = \mathrm{ML}^*(B) = C$, of transcendence degree one, as Theorem A predicts.

6. Remarks on the open cases

Higher dimension. Let $\mathrm{tr.deg}_k B = n \geq 4$ and $2 \leq \mathrm{tr.deg}_k \mathrm{ML}^*(B) \leq n - 2$. The proof of Theorem A still gives the following: if $E \in \mathrm{LND}(B)$ does not kill $R = \mathrm{ML}^*(B)$, then $K = \ker(E|_R)$ contains elements transcendental over k, and no local slice $g \in R$ of $E|_R$ has $E(g)$ invertible in R. Lemma 3.3 gives $R \subseteq \mathrm{ML}(A)$ for every slice kernel A. For $n = 4$ and $\mathrm{tr.deg}_k R = 2$ this says that R is a two-dimensional subring of the three-dimensional ring A killed by every locally nilpotent derivation of A. Over an algebraically closed field the equality $\mathrm{ML}(B) = \mathrm{ML}^*(B)$ is known in all these cases [9]. We do not know whether it holds over $\mathbb{R}$ or $\mathbb{Q}$, and the methods above do not decide it.

Positive characteristic. In characteristic $p > 0$ locally nilpotent derivations are replaced by exponential maps, and ML and ML^* are defined in the same way. Sai Krishna asked whether $\mathrm{ML}^*(B) \neq B$ implies $\mathrm{ML}^*(B) = \mathrm{ML}(B)$ for affine domains over algebraically closed fields of arbitrary characteristic, and answered it in some cases [13]. Our proofs use characteristic zero in Lemma 2.3, where the derivative P' must be nonzero, and in the division by $i!$. They do not apply to that question.

[5] If $p(u)/q(u) = r(v)/s(v)$ with coprime p, q and coprime r, s, then $p(u)s(v) = q(u)r(v)$ in the polynomial ring $\mathbb{C}[u, v]$. Since $q(u)$ is coprime to $p(u)$, it divides $s(v)$, so q is constant; likewise s is constant, and then $p(u)$ is a constant multiple of $r(v)$, so both are constant.

Declarations

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References

[1] I. Arzhantsev, H. Flenner, S. Kaliman, F. Kutzschebauch and M. Zaidenberg, Flexible varieties and automorphism groups, *Duke Math. J.* **162** (2013), no. 4, 767–823. doi:10.1215/00127094-2080132.

[2] D. Bhattacharjee, Non-commutativity over canonical suspension η for genus $g \geq 1$ in hypercomplex structures for potential $\rho\phi$, *Asian Res. J. Math.* **18** (2022), no. 11, 332–341. doi:10.9734/arjom/2022/v18i11607.

[3] D. Bhattacharjee, Establishing equivariant class $[O]$ for hyperbolic groups, *Asian Res. J. Math.* **18** (2022), no. 11, 362–369. doi:10.9734/arjom/2022/v18i11615.

[4] D. Bhattacharjee, S. Singha Roy and P. Nandi, Hodge–toric geometry beyond CY20, *Int. J. Prof. Stud.* **21** (2026), no. 1. doi:10.37648/ijps.v21i01.019.

[5] A. J. Crachiola and L. Makar-Limanov, On the rigidity of small domains, *J. Algebra* **284** (2005), no. 1, 1–12. doi:10.1016/j.jalgebra.2004.09.015.

[6] N. Dasgupta and N. Gupta, An algebraic characterization of the affine three space, *J. Commut. Algebra* **13** (2021), no. 3, 333–345. doi:10.1216/jca.2021.13.333.

[7] A. Dubouloz, The cylinder over the Koras–Russell cubic threefold has a trivial Makar-Limanov invariant, *Transform. Groups* **14** (2009), no. 3, 531–539. doi:10.1007/s00031-009-9051-3.

[8] G. Freudenburg, *Algebraic Theory of Locally Nilpotent Derivations*, 2nd ed., Encyclopaedia Math. Sci. **136**, Springer, Berlin, 2017. doi:10.1007/978-3-662-55350-3.

[9] S. Gaifullin and A. Shafarevich, Modified Makar-Limanov and Derksen invariants, *J. Pure Appl. Algebra* **228** (2024), no. 6, 107616. doi:10.1016/j.jpaa.2024.107616.

[10] N. Gupta, On the cancellation problem for the affine space $\mathbb{A}^3$ in characteristic p, *Invent. Math.* **195** (2014), no. 1, 279–288. doi:10.1007/s00222-013-0455-2.

[11] N. Gupta, On Zariski's cancellation problem in positive characteristic, *Adv. Math.* **264** (2014), 296–307. doi:10.1016/j.aim.2014.07.012.

[12] L. Makar-Limanov, On the hypersurface $x + x^2y + z^2 + t^3 = 0$ in $\mathbb{C}^4$ or a $\mathbb{C}^3$-like threefold which is not $\mathbb{C}^3$, *Israel J. Math.* **96** (1996), no. 2, 419–429. doi:10.1007/BF02937314.

[13] P. M. S. Sai Krishna, An algebraic characterization of the affine three space in arbitrary characteristic, *J. Commut. Algebra* **16** (2024), no. 3, 293–303. doi:10.1216/jca.2024.16.293.

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